

Known answers first:

a certified Landau root, a Vlasov–Poisson solver with its failure modes pinned,
zero sound in Lean, and what persistent homology does and does not add

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SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

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Abstract

We report a small, fully pre-registered validation kit for collisionless damping, spanning a plasma and a Fermi liquid, and we report it including the predictions that failed. (i) The least-damped root of the linearised Vlasov–Poisson dispersion relation for a Maxwellian is enclosed with a computer-assisted certificate in ball arithmetic: at $k\lambda_D = 0.5$, $\omega = 1.4156618886045364\dots - 0.1533594669096048\dots i$, certified to 10^{-40} ; we found no prior certified enclosure. (ii) A semi-Lagrangian solver reproduces the certified rate to 0.14% and distinguishes neighbouring wavenumbers, reproduces the free-streaming recurrence time exactly, and reproduces a closed-form ballistic plasma echo to 2×10^{-9} . (iii) Two predictions that we transferred from free streaming to the self-consistent plasma—the recurrence time and the echo time—*failed* their pre-registered tolerances, both by about 9%, and a resolution study shows the shift is physical. (iv) In Lean 4 we prove that undamped zero sound in a Fermi liquid exists if and only if the Landau parameter is positive, in three dimensions and in two, the latter being the geometry of neutron-scattering experiments on ^3He films; this is textbook physics, newly machine-checked. (v) We use cubical persistent homology strictly as an instrument with known-answer controls. A duality control holds exactly on simulation data. The two instruments built on it *both failed* their pre-registered criteria—a density-only vortex detector (precision 0.87, recall 0.43 against phase-winding ground truth) and a phase-space hole counter—and for one identifiable reason: H_0 persistence is depth relative to a connecting saddle, and in both systems the objects of interest share a valley. Of twenty-five pre-registered criteria and predictions, fourteen were met and eleven were not; three of the eleven were errors in our own criteria or measurement design, and are reported as such.

Nothing here is new physics. Nonlinear Landau damping has just been formalized in Lean by Bedrossian [1]; our Lean results are elementary by comparison and one of them is subsumed by that work. The contribution is the discipline: every number below was preceded by a committed statement of what it should be, and the record of which of those statements were wrong.

1 Introduction

Landau damping of a plasma wave and Landau damping of zero sound in liquid ^3He are one phenomenon: a collective mode of a collisionless kinetic equation loses energy to particles moving at its phase velocity. The first is the standard acceptance test of every Vlasov code; the second decides whether a collective mode of a Fermi liquid is sharp or broad in a neutron-scattering spectrum [2]. This paper builds a small toolkit across that bridge, with one rule: a quantity is computed only after the value it should take, and the tolerance, have been written down and committed. The pre-registration, its three dated amendments, and every script are in the repository.

What is and is not claimed. *New, and small:* a certified enclosure of the Landau root (§2); a machine-checked zero-sound criterion in 2D and 3D (§5); a measured, and poor, error rate for prominence-based vortex detection from density alone, with the mechanism of its failure (§6). *Not new:* the benchmark itself, the plasma echo [3], phase mixing, and—emphatically—anything resembling a theorem about nonlinear Landau damping, which is Mouhot–Villani’s [4] and, in Lean, Bedrossian’s [1].

We began this work believing that a Lean formalization of nonlinear Landau damping was out of reach, and wrote as much in the project’s planning document. The literature check, run before any code, found that it had been posted six days earlier. That planning document now carries an erratum. The episode is the reason §5 is modest, and an instance of this project’s recurring lesson: search first.

2 A certified Landau root

In units $\omega_p = v_{\text{th}} = \lambda_D = 1$, for a Maxwellian background,

$$D(\omega, k) = 1 + \frac{1}{k^2} [1 + \zeta Z(\zeta)], \quad \zeta = \frac{\omega}{\sqrt{2}k}, \quad Z(\zeta) = i\sqrt{\pi} e^{-\zeta^2} \operatorname{erfc}(-i\zeta). \quad (1)$$

Written this way Z is entire, so the formula is already Landau’s analytic continuation to $\operatorname{Im} \omega < 0$ with no case distinction. We evaluate D and D' (using $Z' = -2(1 + \zeta Z)$) in Arb ball arithmetic [5] and apply the interval Newton operator $N(B) = m - D(m)/D'(B)$ to a square ball B with midpoint m : if $N(B) \subset \operatorname{int} B$, then D has exactly one zero in B . All rounding error is inside the balls.

k	ω_r	γ
0.3	1.159846480591913783957252858902818764736	−0.01262036842111715381386333101215427213543
0.4	1.285056969653746093937441697179005657870	−0.06612795869074925181490085255855642353114
0.5	1.415661888604536429791654965787201432819	−0.1533594669096048340930722853059164940159

Table 1: Certified least-damped roots; every digit shown is inside a ball of radius below 3×10^{-40} proved to contain exactly one zero.

The standard numerical reference is Canosa [6], $\gamma = -0.15336$ at $k = 0.5$, with which Table 1 agrees in every digit he gives. A search for a prior computer-assisted enclosure of this root found none; that is an absence of hits, not a proof of absence. The same code certifies the purely growing root of the two-stream background used in §6.

Controls. A ball of radius 10^{-3} about $1.4156 - 0.1433i$ —the right frequency with the damping rate off by 0.01—is *not* certified; the same ball about the true root is.

What is certified. A zero of the formula above. That the formula describes the physics is not something ball arithmetic can know; it is tested against an independent solver in §3.

A failed criterion. Our pre-registered pass condition was that the certified midpoint “rounds to” the literature’s 1.4156 and -0.1533 . It rounds to 1.4157 and -0.1534 : the literature values are truncations. By the letter the criterion failed, through our wording. We record it rather than reword it.

3 A solver validated against the certificate

The solver is deliberately ordinary: Strang splitting, exact Fourier shift in x , cubic-spline semi-Lagrangian advection in v , FFT Poisson solve. Parameters were fixed in advance ($N_x = 64$, $N_v = 256$, $v_{\max} = 6$, $\Delta t = 0.1$, perturbation $0.01 \cos kx$, rate from a line through the maxima of $\ln |E_1|$ in $5 \leq t \leq 25$).

run	k	measured γ	error	measured ω	error
A	0.5	-0.15358	0.14%	1.41521	0.03%
B	0.4	-0.06637	0.36%	1.28485	0.02%

Table 2: Against the certified roots; tolerances were 2% and 1%. Run B misses the $k = 0.5$ root by 57%: the comparison can tell wavenumbers apart, which is what makes Run A informative. Mass drifts by 7×10^{-10} over $t = 50$, which *fails* our pre-registered 10^{-10} ; we believe, but have not verified, that this is the zero-fill beyond $\pm v_{\max}$ in the spline step. Energy drift, for which we set no threshold, is below 6×10^{-7} .

Recurrence. The velocity quadrature aliases the free-streaming phase e^{-ikvt} at $T_R = 2\pi/(k\Delta v)$. We pre-registered that at $N_v = 32$ ($T_R = 33.51$) the largest maximum of $|E_1|$ in $25 < t < 45$ would fall within 5% of T_R .

Failed: 36.5, an error of 8.9%. A diagnostic written *after* the failure, and labelled as such in the repository, shows why. With the field switched off, the density mode recurs at $t = 33.50$ with amplitude ratio 1.000: the formula is exact for what it describes. With the field on, the recurring perturbation drives a Langmuir oscillation and the field's largest maximum lags by 3.0, about two thirds of an oscillation period. We had applied a free-streaming formula to a self-consistent quantity.

4 The plasma echo

A perturbation in mode k_1 at $t = 0$ phase-mixes away; a second in mode k_2 at $t = \tau$ does the same; at second order their product contains a term whose velocity phase unwinds at

$$t_e = \tau \frac{k_2}{k_2 - k_1}, \quad (2)$$

producing a macroscopic response in mode $k_2 - k_1$ from two disturbances that have both disappeared [3]. Controlling the accumulation of such echoes is the central difficulty of the nonlinear theory [4]. For free streaming with multiplicative pulses of amplitude α_1, α_2 the response is closed-form,

$$\rho_{k_2-k_1}(t) = \frac{1}{2} \alpha_1 \alpha_2 \exp\left(-\frac{1}{2}[(k_2 - k_1)t - k_2\tau]^2\right), \quad (3)$$

the Gaussian being the transform of the Maxwellian, which we also prove in Lean (`maxwellian_mode`, §5). With $k_1 = 0.5$, $k_2 = 1.5$, $\tau = 10$, so $t_e = 15$:

- **Field off.** The solver matches (3), prefactor included, to 2×10^{-9} of the peak (tolerance 10^{-3}), peaking at $t = 15.00$. Without the second pulse the echo mode stays at 10^{-12} of the peak.
- **Field on.** We predicted a peak within 5% of 15.0.

Failed: 16.40, an error of 9.3%, with amplitude 2.08 times the ballistic value. Doubling the velocity resolution and halving the time step gives 16.400 again and the same amplitude to 0.04%, so the shift is not numerical: the self-consistent field delays and amplifies the echo. The field-on negative control also failed its 1% threshold (1.5%): a single pulse generates its own second harmonic at order α_1^2 , which we had not allowed for. Both are known physics—the echo’s shape depends on the dielectric response—that we did not apply when writing the prediction.

The pattern. Every exact known answer was reproduced to near rounding error. Both predictions that failed were free-streaming formulas we had transferred to the interacting system, and both were off by about 9% at these parameters. For a validation suite the practical conclusion is to test recurrence and echo timing *with the field off*, where they are exact, and to treat the field-on values as measurements.

5 Machine-checked statements

All in Lean 4.34.0-rc2 with Mathlib, standard axioms only, each with a perturbed statement that is required to fail, inside a library of 145 theorems described elsewhere [7].

Zero sound. With a single Landau parameter $F = F_0^s$, a collective mode of phase velocity $s v_F$ satisfies $1 + F \Omega(s) = 0$; it is undamped iff $s > 1$, outside the particle–hole continuum, and Landau-damped otherwise [8]. In three dimensions $\Omega(s) = 1 - \frac{s}{2} \ln \frac{s+1}{s-1}$; in two, $\Omega(s) = 1 - s/\sqrt{s^2 - 1}$.

```
theorem zero_sound_iff (F : ℝ) :
  (exists s, 1 < s /\ s / 2 * log ((s + 1) / (s - 1)) - 1 = 1 / F) <-> 0 < F
theorem zero_sound_2d_iff (F : ℝ) :
  (exists s, 1 < s /\ 1 + F * (1 - s / sqrt (s ^ 2 - 1)) = 0) <-> 0 < F
```

(ASCII rendering.) The 3D proof brackets $g(s) = \frac{s}{2} \ln \frac{s+1}{s-1} - 1$ between $\frac{1}{2} \ln \frac{2}{s-1} - 1$ and $1/(s-1)$, using Mathlib’s $\frac{2x}{x+2} < \ln(1+x)$ for positivity, and closes with the intermediate value theorem. In 2D the root is explicit, $s = (1 + F)/\sqrt{1 + 2F}$. Replacing $0 < F$ by $0 \leq F$ fails to compile in both. The 2D case is included because the neutron-scattering observation of a collective mode in a Fermi liquid by Godfrin et al. [2] is on a monolayer, where the logarithmic formula does not apply. Uniqueness of the 3D root, F_1^s , finite temperature and the damped branch are not treated.

Phase mixing. For any $g : \mathbb{R} \rightarrow \mathbb{C}$ and $k \neq 0$, $\int g(v) e^{-ikvt} dv \rightarrow 0$; $e^{ik(x-vt)} g(v)$ solves the transport equation; and for the unit Maxwellian the mode is exactly $e^{-k^2 t^2/2}$. The first is a reparametrisation of Mathlib’s Riemann–Lebesgue lemma and is subsumed by [1]; it is here because (3) is a known answer, and a known answer should be a checked one.

6 Persistent homology as an instrument

This project has previously made three topological proposals about quantum fluids; one was refuted by simulation, one withdrawn, and one stopped at the literature check as a renaming of topographic prominence [7]. We therefore framed this section as instruments with known-answer controls, and state at the outset what the central quantity is: the H_0 sublevel bar of a local minimum is (value at the minimum, value at the saddle where its basin joins a deeper one). Its length is prominence, upside down. Nothing below is new mathematics. All computations use GUDHI [9] cubical complexes.

6.1 A duality control, and why it matters

For a function on a closed surface, finite H_0 sublevel pairs (b, d) of ρ correspond to finite H_1 sublevel pairs $(-d, -b)$ of $-\rho$ [10]. On pixel data this holds exactly only if the two sides use *dual* cubical constructions—pixels as top-dimensional cells on one side, as vertices on the other—and with periodic identification these are dual filtrations of the torus [11]. We pre-registered both the agreement and its converse: with the *same* construction on both sides the diagrams must differ, since the two sides then use different pixel connectivities.

field	dual constructions	same construction
smooth random, 128^2	81 vs 81 pairs, max difference 0	81 vs 116 pairs
$ \psi ^2$, Gross–Pitaevskii, 1024^2	985 vs 985 pairs, max difference 0	985 vs 1772 pairs

Table 3: The duality control. Exact agreement, and exact disagreement where predicted.

A first version of our check that GUDHI’s cell indices map to the right pixels compared a value with itself through the same indexing and could not fail; it was replaced by the test that every reported birth pixel is a local minimum (986/986). The control is not decoration: it says that the H_1 superlevel diagram carries the same numbers as the H_0 sublevel one, so when the instruments below failed, changing homological degree was already known not to be a remedy.

6.2 Vortices from density alone: refuted in compressible turbulence

Experiments on condensates image $|\psi|^2$, not the phase, so the winding number that defines a vortex is not available; vortices are counted by blob detection or learned detectors [12], and persistent homology has been applied to condensate densities for other purposes [13, 14]. We asked what a threshold-on-prominence detector achieves against winding-number ground truth, on our own two-dimensional Gross–Pitaevskii decaying turbulence ($\xi/\Delta x = 8$, frames at $t = 5, 10, 20$, about 830 vortices). Fixed in advance: a detection is a minimum with value below $0.1 n_0$ and depth above $0.5 n_0$, matched one-to-one within ξ ; “usable” meant precision ≥ 0.95 and recall ≥ 0.90 .

Failed: precision 0.871, recall 0.433. We had predicted that recall would fail only for close pairs, and would be at least 0.97 for vortices with no neighbour within 2ξ . It is 0.51. Isolation in position does not imply isolation in the sublevel-set topology.

A diagnostic written afterwards locates the cause. At $t = 10$, *all* 289 vortices have a density minimum below $0.1 n_0$ within ξ . Their depths are below 0.1 for 62, 0.1–0.3 for 52, 0.3–0.5 for 46, and above 0.5 for only 129. The state is strongly compressible—density standard deviation $0.45 n_0$, 15.5% of the area below $0.5 n_0$ —so cores are joined by low-density channels and lose their prominence; and 142 of the 462 near-zero minima belong to no vortex. No pair of thresholds on our sensitivity grid exceeds recall 0.58. The negative control passed (a vortex-free frame with density between 0.18 and $1.60 n_0$: no detections among 506 minima). Whether the detector works in a quiet condensate, the experimentally relevant case, we did not test and do not claim.

6.3 Phase-space holes: refuted

For the two-stream instability ($f_0 = \frac{1}{2}[M(v - 2.4) + M(v + 2.4)]$, $k = 0.2$) the purely growing root is certified as in §2: $\gamma = 0.22584425503471 \dots$

Failed: measured growth rate 0.18656, an error of 17% (tolerance 5%). Either the certificate or the measurement was wrong, so we tested both. Direct quadrature of $D(i\gamma)$, valid without continuation for $\text{Im } \omega > 0$, gives 8×10^{-15} at the certified root and -0.54 at the measured value. The pre-registered seed of 10^{-3} left only two decades below saturation, and the local growth rate slides from 0.33 to 0.21 without a plateau: there was no linear phase to fit. With a seed of 10^{-8} the rate plateaus at 0.22591, 0.22585, and a fit gives 0.22594, within 0.04% of the certificate. The solver was right and our measurement design was not.

We then counted holes as H_0 classes born at $|v| < 2.4$ with depth above $0.1 \max f$ (the treatment of the essential class was fixed by a dated amendment before any two-stream data existed). Predicted: one hole at $t = 60$ with one wavelength in the box; two at saturation with two.

Both failed. With one wavelength the count is 1 until saturation, then 8, 29, 14, 52 at $t = 50, \dots, 80$: the trapped region filaments, the filaments are deep, and a depth threshold does not tell a filament from a hole. With two wavelengths the two physically equivalent holes score depths 0.84 and 0.17 at saturation: the beams leave a trough at $v \approx 0$ where f_0 is 11% of its maximum, the holes are joined through it, and by the elder rule only one keeps a long bar. The negative control (a stable Maxwellian) correctly gave zero.

One mechanism. Both instruments failed the same way. Persistence measures depth relative to the *connecting saddle*; vortex cores in compressible turbulence and phase-space holes between two beams both sit in a shared valley, where that saddle is low. By the duality control the other homological degree reports the same numbers. For these two problems prominence is the wrong discriminator, and it is the geometry of the field, not a choice of threshold, that makes it so.

7 Discussion

The scorecard. Twenty-five criteria and predictions were committed before the corresponding code existed. Fourteen were met: the certificate and its control; both Landau runs, with the discrimination test; the ballistic echo and its field-off control; the three Lean statements; both halves of the duality control; one sub-prediction about close vortex pairs; and both topological negative controls. Eleven were not. Three were our own errors of design: a rounding criterion written against truncated literature values, a seed too large to leave a linear phase, and a negative-control threshold that ignored self-generated harmonics. One, mass conservation at 7×10^{-10} against a threshold of 10^{-10} , we cannot yet account for. Two were free-streaming formulas misapplied to the interacting plasma. Five were wrong predictions about what persistence would detect.

What we take from it. First, the exact known answers—certified root, ballistic echo, free-streaming recurrence—were all reproduced at or near rounding error, and are the tests we recommend; the self-consistent echo and recurrence times should be treated as measurements, not checks. Second, a certificate earns its keep when a measurement disagrees with it: the 17% two-stream discrepancy was resolved in minutes because one side of the comparison could not be wrong in the way the other could, and an independent quadrature confirmed which. Third, persistent homology of a scalar field is prominence, and prominence fails wherever the features of interest share a valley; that is a statement about two specific configurations, not about topological data analysis.

What this is for. The kit is offered as validation infrastructure for anyone writing a Vlasov code or analysing collective modes in a Fermi liquid. It makes no claim on the mathematics of Landau damping, whose nonlinear theory [4] now has a formal proof [1]; the dispersion relation and Penrose criterion, which that formalization explicitly leaves aside, are where certified numerics of the present kind might eventually meet it.

Process note. All predictions, analysis, code and text were produced with an AI assistant under the author’s direction; the pre-registration discipline exists precisely because such a process generates plausible claims faster than they can be checked by reading.

Availability. Code, pre-registration with amendments, run scripts and outputs: <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids> (src/quantumfluids/kinetic, src/quantumfluids/tda/cubical.py, docs/designs/KINETIC_TDA_PREREG.md).

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